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Practical No. 04

Aim: Create a HTML web page that checks whether the input number is even or odd using JavaScript and display the result dynamically using DOM..

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AIM : Create a HTML web page that checks whether the input number is even or odd using JavaScript and display the result dynamically using DOM..

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- Able to analyze various registration forms.
- To understand the concept of the DOM tree.
- To understand the concept of DOM.

HARDWARE AND SOFTWARE REQUIRMENTS:

Hardware Requirement: Modern computer, Memory, processor etc.

Software Requirement: Github, VS code, online compiler etc.

THEORY:

The Document Object Model (DOM) is a programming API for HTML and XML documents. It defines the logical structure of documents and the way a document is accessed and manipulated. In the DOM specification, the term "document" is used in the broad sense - increasingly, XML is being used as a way of representing many different kinds of information that may be stored in diverse systems, and much of this would traditionally be seen as data rather than as documents. Nevertheless, XML presents this data as documents, and the DOM may be used to manage this data.

With the Document Object Model, programmers can create and build documents, navigate their structure, and add, modify, or delete elements and content. Anything found in an HTML or XML document can be accessed, changed, deleted, or added using the Document Object Model, with a few exceptions - in particular, the DOM interfaces for the internal subset and external subset have not yet been specified.

In the Document Object Model, documents have a logical structure which is very much like a

tree; to be more precise, it is like a "forest" or "grove" which can contain more than one tree. However, the Document Object Model does not specify that documents be *implemented* as a tree or a grove , nor does it specify how the relationships among objects be implemented in any way. In other words, the object model specifies the logical model for the programming interface, and this logical model may be implemented in any way that a particular implementation finds convenient. In this specification, we use the term *structure model* to describe the tree-like representation of a document; we specifically avoid terms like "tree" or "grove" in order to avoid

implying a particular implementation. One important property of DOM structure models is *structural isomorphism*: if any two Document Object Model implementations are used to create a representation of the same document, they will create the same structure model, with precisely the same objects and relationships.

CODE:

```
<html>
    <head>
      <h1>Find Even and Odd number </h1>
    </head>
    <body>
<h2> Debjit Talukdar  CS24202</h2>
      <h3>Find Even and Odd among the number
<label>Enter the number: </label>
      <input type="number" id="num"><br><br>
      <button onclick="checkEvenOdd()">Submit</button>
      <p id="result"></p>
      <script>
      function checkEvenOdd
        const number = parseInt(document.getElementById("num").value);
        if(number % 2 === 0) {
          document.getElementById("result").innerText = number + " is even ";
        }else{
          document.getElementById("result").innerText = number + " is odd ";
        }
      } </script>
</body>
</html>
```

OUTPUT

CONCLUSION: Hence by doing this practical and program we are able to Create a HTML web page that checks whether the input number is even or odd using JavaScript and display the result dynamically using DOM..

DISCUSSION AND VIVA VOCE:

1. What are the properties of HTML DOM?
2. What are the HTML DOM methods involved?
3. Explain the relationship between the node, child, and sibling by example.
4. What is an event and how can you define an event in HTML DOM?
5. How many types of events in HTML DOM?

REFERENCE:

- https://www.w3schools.com/
- https://www.geeksforgeeks.org/dom-document-object-mode
- https://www.typescriptlang.org/docs/handbook/dom-manipulation.html